GOVERNMENT HEALTH INFRASTRUCTURE ANALYSIS — INDIA: A DISTRICT-WISE STUDY OF FACILITY DISTRIBUTION, REGIONAL DISPARITIES, AND PREDICTIVE MODELING

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Abstract—*This paper presents a comprehensive data-driven analysis of government healthcare infrastructure across Indian states and districts using the Rural Health Statistics dataset (2016–2021). The study examines the distribution of Sub Centres (SCs), Primary Health Centres (PHCs), Community Health Centres (CHCs), Sub Divisional Hospitals, and District Hospitals across 731 districts in 36 states and union territories. Through multi-phase data preprocessing, exploratory data analysis (EDA), visualisation, correlation analysis, and a linear regression predictive model, this work identifies significant regional disparities, underserved regions, dominant facility types, and year-wise growth patterns. The predictive model achieves an R^2 score of 0.8286 and a Mean Absolute Error (MAE) of 1.92, demonstrating that healthcare infrastructure follows predictable, learnable patterns that can support data-driven policy and resource allocation.*

Index Terms—*Healthcare Infrastructure, Rural Health Statistics, India, Data Analysis, Linear Regression, PHC, CHC, Sub Centres, Regional Disparities, Predictive Modeling.*

I. INTRODUCTION

Access to timely and quality healthcare is a fundamental human right. The geographic distribution of healthcare facilities directly governs the health outcomes of populations, particularly in developing nations where spatial inequalities are pronounced. India operates a structured, multi-tier public healthcare delivery system that spans from grassroots Sub Centres serving villages of 3,000–5,000 people, through Primary Health Centres (PHCs) covering groups of villages, to Community Health Centres (CHCs) providing specialist care, and District Hospitals offering advanced referral services.

Despite sustained government investment through schemes such as the National Health Mission (NHM), wide disparities persist across states, regions, and districts. Some districts boast hundreds of PHCs while others barely reach double digits. These gaps translate directly into unequal healthcare access and avoidable mortality. Understanding the spatial distribution and growth patterns of these facilities is therefore a prerequisite for effective policy planning.

This study analyses the Rural Health Statistics Districtwise Healthcare Infrastructure dataset published by India's Ministry of Health and Family Welfare, covering financial years 2016 through 2021. With 3,817 clean records spanning 36 states and union territories and 731 districts, the dataset provides a rich longitudinal view of facility availability at the district level. This paper contributes: (1) a documented data cleaning and preprocessing pipeline; (2) district- and state-level facility distribution analyses with visualisations; (3) temporal growth trend analysis; (4) inter-facility correlation analysis; and (5) a linear regression model predicting CHC availability from related infrastructure features.

The importance of this analysis extends beyond academic interest. India's National Health Policy 2017 set ambitious targets including universal access to free essential health services, reduction of infant and maternal mortality, and a fourfold increase in health expenditure as a share of GDP. Meeting these targets requires granular understanding of existing infrastructure gaps at the district level — precisely the kind of insight that systematic data analysis of the Rural Health Statistics dataset can provide.

Previous studies on India's healthcare infrastructure have largely focused on state-level comparisons or macro-level policy assessments. District-level data-driven analyses remain comparatively scarce in the published literature. This paper addresses that gap by combining thorough data engineering, multi-dimensional exploratory analysis, and a predictive modelling exercise to generate actionable, district-level intelligence about healthcare facility distribution and growth.

II. DATASET AND METHODOLOGY

A. Data Source

The primary dataset is the Rural Health Statistics Districtwise Health Care Infrastructure file, obtained from the Government of India data portal. It contains 3,874 raw entries covering eleven columns: Country, State, District, Year, Functional Sub Centres, Functional PHCs, Functional CHCs, Functional HWC-SCs, Functional HWC-PHCs, Functional Sub Divisional Hospitals, and Functional District Hospitals. The dataset covers the period from the financial year 2016 (April–March) through 2021.

B. Data Cleaning — Phase 1

Column names contained trailing whitespace and were verbose. Each column was stripped and renamed to short, readable identifiers. Columns reporting Health and Wellness Centres (HWC-SCs and HWC-PHCs) contained over 83% missing values and were dropped using a 50% completeness threshold, reducing the schema to nine columns. This step improved downstream readability and eliminated noise from sparsely populated fields.

C. Data Preprocessing — Phase 2

Duplicate rows were identified and removed. Rows where all values were null were dropped. Textual fields (State, District) were stripped of leading and trailing whitespace. Entries with invalid or placeholder district names such as 'Unknown Districts of India' and single-word regional placeholders were filtered out. The Year column was parsed from its original verbose string format ('Financial Year (Apr – Mar), 2021') using a regex extractor and converted to integer. All facility count columns were cast to numeric dtype, with residual null values replaced by zero and subsequently cast to integer. After all cleaning and filtering operations, 3,817 valid records remained, representing a loss of only 1.5% of original entries.

D. Analysis and Modeling — Phases 3 & 4

Exploratory analysis was conducted on the most recent year (2021) for cross-sectional comparisons. Year-wise aggregation was used for longitudinal trend analysis. A scikit-learn Pipeline combining OneHotEncoder (for the categorical State feature) with LinearRegression was trained to predict Functional CHC counts using Sub Centres, PHCs, Year, and State as inputs. The data was split 80/20 for training and testing with a fixed random seed (42) to ensure reproducibility.

TABLE I

Key Dataset and Model Statistics

Metric	Value
Total Records	3,817
States / UTs	36
Districts	731
Year Range	2016 – 2021
Avg. Sub Centres / District	236
Max Sub Centres (single district)	19,419
Avg. PHCs / District	40.4
Avg. CHCs / District	8.6
Linear Regression R^2	0.8286
Linear Regression MAE	1.92 CHCs

III. RESULTS AND ANALYSIS

A. Top Districts by Primary Health Centres

Figure 1 presents the top five districts ranked by functional PHC count in 2021. Bengaluru Urban leads with 203 PHCs, followed by Pune (192), Jaipur (182), East Godavari (163), and Guntur (163). This concentration of PHCs in urban and semi-urban districts reflects the compounding effect of higher population densities, better administrative capacity, and historically stronger investment in major cities. Rural and tribal districts, by contrast, often struggle to maintain even the minimum norm of one PHC per 30,000 rural population.

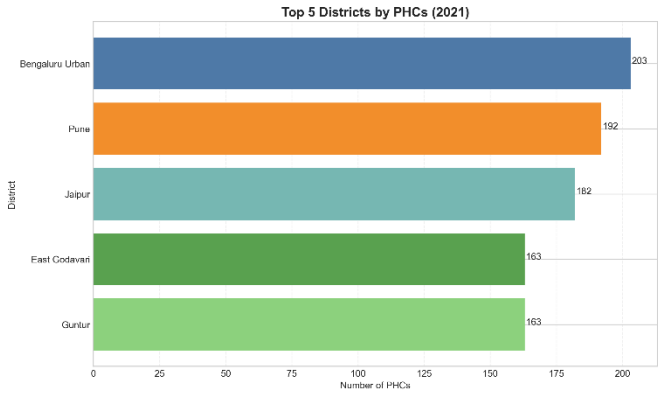

Fig. 1. Top 5 Districts by Functional PHCs (2021).

B. Top Districts by Community Health Centres

For CHCs, Alwar leads with 42 units, tied closely by Jaipur (41), Pune (41), Nagaur (34), and Udaipur (33). The overlap between districts topping both PHC and CHC rankings points to a pattern of co-located infrastructure investment. Districts that have successfully scaled their primary care network appear to attract secondary-level facility development as well, while districts lagging in PHCs are almost universally also deficient in CHCs.

C. Top 10 States by Overall Infrastructure

Figure 2 shows a grouped bar chart comparing the top 10 states by total facility count across Sub Centres, PHCs, and CHCs for 2021. Uttar Pradesh dominates Sub Centre volumes with close to 20,000 facilities, followed by Rajasthan and Andhra Pradesh. In all states, Sub Centres vastly outnumber PHCs and CHCs, maintaining the classic pyramid structure of India's tiered health system. States such as Maharashtra and West Bengal show more balanced distributions across all three tiers, suggesting a more mature infrastructure development trajectory.

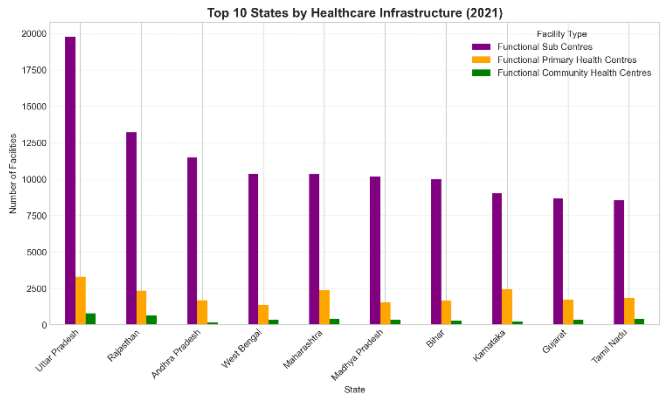

Fig. 2. Top 10 States by Healthcare Infrastructure (2021).

D. Year-wise Growth Trends

Figure 3 illustrates the average number of facilities per district from 2016 to 2021. Sub Centre averages declined slightly from 2016 to 2018 before recovering to approximately 230 by 2021. PHC averages held relatively steady at around 40 per district throughout the period. CHC averages remained the lowest, near 8 per district, reflecting the limited expansion of secondary-level rural infrastructure over this period. The dip in 2017–2018 may reflect data reporting lags or rationalisation of facility counts during NHM reform cycles.

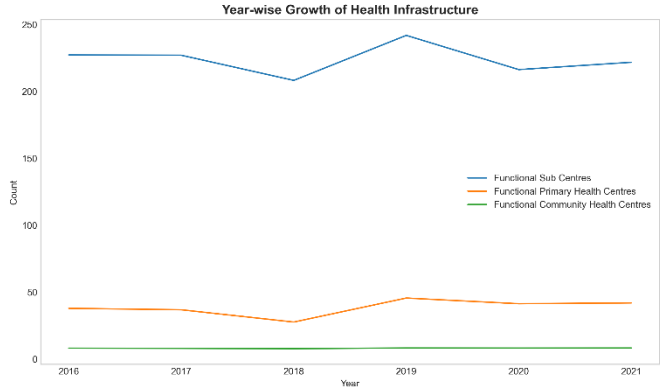

Fig. 3. Year-wise Average Growth of Healthcare Facilities (2016–2021).

E. Distribution of Facility Types

Figure 4 presents the proportional distribution of all healthcare facilities in 2021. Sub Centres account for 81.5% of total facilities, underscoring their dominant role as the first point of contact for rural healthcare. PHCs contribute 15.4%, while CHCs constitute just 3.1%. This steep pyramid indicates that while primary access points are broadly distributed, the density of higher-order facilities capable of managing complicated cases and surgical procedures remains critically low in relation to India's population.

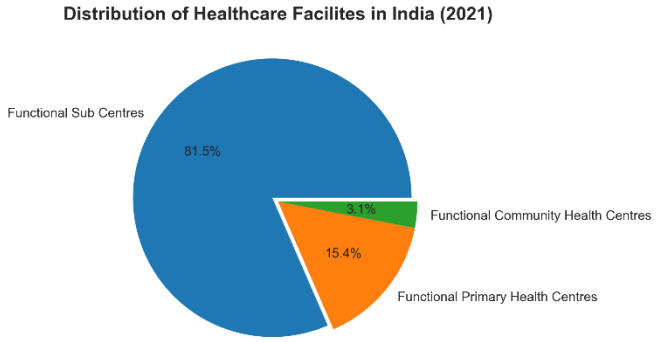

Fig. 4. Proportional Distribution of Healthcare Facilities in India (2021).

F. Top Districts by Sub Centres

Among Sub Centres, East Godavari (Andhra Pradesh) leads with 1,288, followed by Chittoor (1,070), South 24 Parganas (West Bengal, 1,068), West Godavari (940), and Kurnool (926), as shown in Figure 5. These districts correspond to high-density rural populations where Sub Centres serve as the exclusive first point of medical contact. Their high counts reflect both demographic demand and, in the Andhra Pradesh districts, historically strong Sub Centre deployment under state rural health schemes.

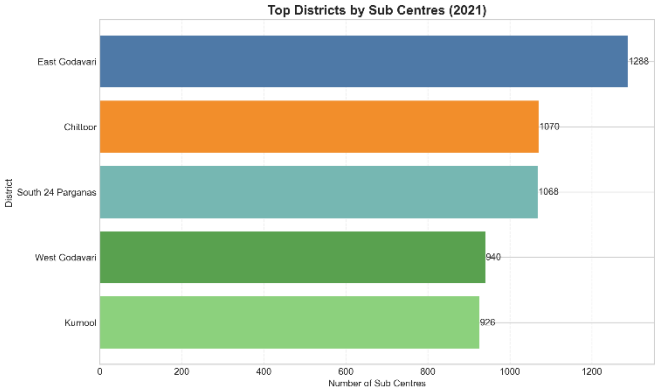

Fig. 5. Top 5 Districts by Functional Sub Centres (2021).

IV. CORRELATION ANALYSIS

Figure 6 presents the Pearson correlation matrix for the four main facility types. Strong positive correlations are observed between Sub Centres and PHCs (0.79), Sub Centres and CHCs (0.78), and PHCs and CHCs (0.71). These high values confirm that the three primary and secondary facility categories tend to grow together across districts. Regions that have invested in Sub Centres generally also have denser PHC and CHC networks, suggesting a coordinated or cumulative model of healthcare infrastructure development.

Conversely, District Hospitals show weak correlations with all other facility types: 0.16 with Sub Centres, 0.12 with PHCs, and 0.21 with CHCs. This divergence is consistent with the hypothesis that District Hospital placement is driven primarily

by administrative and political factors — one hospital per revenue district as a policy norm — rather than by demand derived from lower-tier facility densities. The implication for policy is that strengthening the referral chain requires deliberate effort to align hospital placement with actual catchment infrastructure.

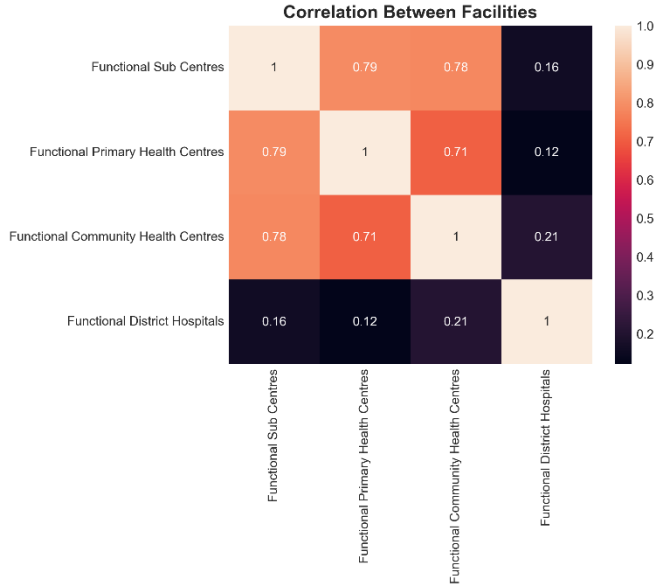

Fig. 6. Pearson Correlation Heatmap of Functional Healthcare Facilities (2021).

V. PREDICTIVE MODELING

A supervised linear regression model was constructed to estimate the number of Functional CHCs per district given Sub Centre count, PHC count, Year, and State. The State variable was one-hot encoded, capturing state-level fixed effects that reflect policy environment, administrative capacity, and historical investment patterns. The model was trained on 80% of the data (3,053 records) and evaluated on the remaining 20% (764 records).

The model achieved the following performance metrics on the held-out test set:

- Mean Absolute Error (MAE): 1.92 CHCs per district
- R^2 (Coefficient of Determination): 0.8286

These results indicate that over 82% of variance in CHC counts is explained by the selected features. An average prediction error of fewer than 2 CHCs per district is clinically meaningful given that most districts have between 3 and 15 CHCs. Figure 7 shows the scatter plot of predicted versus actual CHC values; the majority of points cluster tightly along the perfect-prediction diagonal, with wider dispersion only at higher CHC counts where data is sparser.

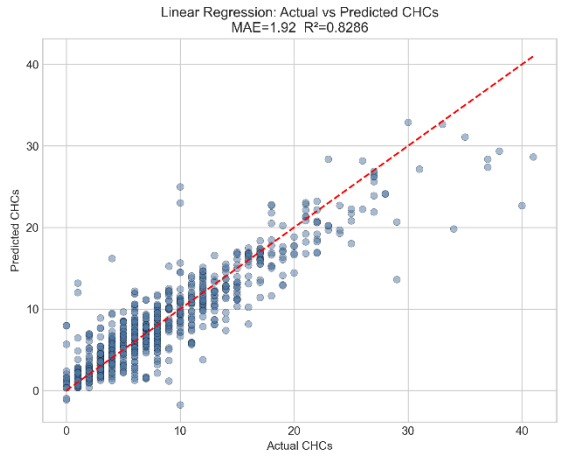

Fig. 7. Actual vs. Predicted CHCs — Linear Regression (R^2=0.8286, MAE=1.92).

Analysis of the numeric feature coefficients shows that PHC count carries the largest positive weight, followed by Sub Centre count, and then Year. The positive Year coefficient indicates a marginal year-on-year increase in predicted CHCs even when facility counts are held constant, reflecting secular expansion trends. Including State as a categorical predictor substantially improves model fit by absorbing inter-state heterogeneity that numeric features alone cannot capture.

VI. DISCUSSION

The analysis reveals several structurally important patterns in India's rural healthcare infrastructure that carry direct policy implications.

Geographic Concentration: Healthcare resources are heavily concentrated in urban and semi-urban districts. Bengaluru Urban, Pune, and Jaipur consistently lead PHC and CHC rankings, while hundreds of smaller districts remain chronically underserved. Correcting this imbalance requires targeted investment guided by population-adjusted metrics rather than absolute facility counts.

Pyramidal Structure and Secondary Care Deficit: Sub Centres account for over 80% of all facilities, while CHCs represent only 3.1%. This extreme pyramid means that patients requiring secondary care must often travel long distances or seek expensive private alternatives. Expanding CHC capacity in regions with high Sub Centre and PHC densities — where demand is demonstrably present — would yield the highest returns in improved health access.

Co-development of Primary Facilities: The strong intercorrelations (0.71–0.79) among Sub Centres, PHCs, and CHCs suggest bundled infrastructure development. Districts that have built out Sub Centres tend to attract PHC and CHC investment. Policy levers that accelerate Sub Centre expansion

in lagging districts may therefore catalyse broader infrastructure improvement over time.

Temporal Stagnation in Higher-order Facilities: The year-wise trend data shows that while Sub Centre averages have grown modestly, PHC and CHC averages have remained nearly flat over the 2016–2021 period. Sustained growth in secondary and tertiary facilities requires dedicated budget allocations beyond what organic infrastructure co-development provides.

Predictive Utility: The linear regression model's R^2 of 0.8286 validates the hypothesis that CHC distribution is largely predictable from primary facility features and state identity. This opens the door to proactive gap identification: policymakers can run the model on proposed Sub Centre and PHC expansion plans to forecast expected CHC requirements, enabling pre-emptive resource planning rather than reactive responses to identified deficits.

VII. CONCLUSION

This paper has presented a reproducible, multi-phase data analysis of India's government healthcare infrastructure at the district level for 2016–2021. Starting from a raw 3,874-record dataset, a rigorous cleaning and preprocessing pipeline produced 3,817 analysis-ready records. Exploratory analysis and visualisation revealed pronounced regional disparities, a pyramidal facility structure dominated by Sub Centres, and broadly stagnant growth in secondary-care facilities over the study period. Correlation analysis confirmed strong co-development among primary facility tiers and highlighted the structural disconnect of District Hospital placement. A linear regression model predicting CHC counts from Sub Centres, PHCs, Year, and State achieved an R^2 of 0.8286 and an MAE of 1.92, demonstrating high predictive utility for infrastructure planning.

The findings collectively underscore the urgent need for data-driven healthcare planning policies that prioritise geographic equity, accelerate secondary facility expansion in underserved districts, and align hospital placement with the primary infrastructure catchment network. Future research directions include integrating population census data for per-capita adjusted analyses, adding road-network and urbanisation variables to improve model accuracy, applying non-linear models such as Random Forest or Gradient Boosting for comparison, and developing interactive geospatial dashboards to make these insights accessible to district health planners.

To maximise policy impact, the outputs of this kind of analysis should be institutionalised within state and district health planning processes. Integrating data pipelines that automatically update facility gap analyses as new Rural Health Statistics releases are published would enable near-real-time monitoring of infrastructure equity. Partnerships between health ministries, academic institutions, and open-data platforms can lower the barrier to such evidence-based governance and help ensure that every Indian citizen, regardless of geographic location, has equitable access to life-saving healthcare services.

VIII. LIMITATIONS AND FUTURE WORK

This study carries several limitations that should be acknowledged. First, the dataset does not include population figures at the district level, preventing calculation of per-capita facility ratios, which would be a more equitable measure of healthcare access than raw facility counts. Future work should merge this dataset with Census 2011 or SECC data to compute population-adjusted metrics and identify districts that are statistically underserved relative to their population size.

Second, the analysis is restricted to functional facility counts and does not capture facility quality, staffing levels, equipment availability, or actual utilisation rates. A district may have a formally registered PHC that operates with a single nurse and no physician, yielding very different health outcomes than a fully staffed facility. Integrating quality and human resource data from HMIS reports would significantly enrich the policy relevance of the analysis.

Third, geographic accessibility is not captured. A district may have an adequate number of CHCs but concentrate them in urban pockets, leaving rural sub-populations effectively without access. Geospatial analysis incorporating road networks, travel-time isochrones, and population distribution would address this gap. Fourth, the predictive model is limited to linear regression. Ensemble methods such as Random Forests and Gradient Boosted Trees may capture non-linear state and district-level interactions more effectively and should be explored in follow-up studies.

IX. REFERENCES

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